

Simulation study: mixture of experts model

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Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 3$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.35, 0.40, 0.25)$
- Degrees of freedom $\boldsymbol{\nu} = (8, 12, 3)$
- Scale matrices of the Wishart distribution $\Sigma_1, \Sigma_2, \Sigma_3$
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2) + \pi_3 \text{Wishart}(\nu_3, \Sigma_3)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(mewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.35, 0.40, 0.25),
  nus = c(8, 12, 3),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 3

# run a mewishart model
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
fit <- mewishart(S_list,
  K = my_K_intial, init_pi = c(0.45, 0.40, 0.15),
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)

Iter    1 | LL=-1913.4 | 60.1 iter/sec
Iter  500 | LL=-1842.1 | 869.9 iter/sec
Iter 1000 | LL=-1842.1 | 903.6 iter/sec
Iter 1500 | LL=-1839.9 | 910.1 iter/sec
Iter 2000 | LL=-1837.4 | 919.3 iter/sec
```

```

Iter 2500 | LL=-1843.7 | 925.6 iter/sec
Iter 3000 | LL=-1840.5 | 927.5 iter/sec
Iter 3500 | LL=-1846.2 | 927.5 iter/sec
Iter 4000 | LL=-1841.4 | 930.0 iter/sec
Iter 4500 | LL=-1841.1 | 932.5 iter/sec
Iter 5000 | LL=-1838.3 | 934.3 iter/sec
Iter 5500 | LL=-1839.2 | 934.9 iter/sec
Iter 6000 | LL=-1840.3 | 935.1 iter/sec
Iter 6500 | LL=-1844.6 | 936.2 iter/sec
Iter 7000 | LL=-1839.0 | 937.0 iter/sec
Iter 7500 | LL=-1843.5 | 935.9 iter/sec
Iter 8000 | LL=-1840.1 | 936.9 iter/sec
Iter 8500 | LL=-1830.8 | 937.8 iter/sec
Iter 9000 | LL=-1832.9 | 937.5 iter/sec
Iter 9500 | LL=-1836.0 | 937.8 iter/sec
Iter 10000 | LL=-1828.3 | 938.2 iter/sec

```

```

end_time <- Sys.time()
elapsed_time <- end_time - start_time
elapsed_time

```

Time difference of 10.6677 secs

```

# posterior summaries
colMeans(fit$pi) # estimated pi

```

```
[1] 0.44828623 0.09349135 0.45822242
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit$nu, 2, mean) # estimated nu
```

```
[1] 11.232330 9.622642 5.062156
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```

# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean <- Reduce("+", fit$Sigma) / length(fit$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean[, , 1])

```

```

      [,1]      [,2]
[1,] 2.093033 0.462634
[2,] 0.462634 1.535057

```

```
print(sigma_posterior_mean[, , 2])
```

```

      [,1]      [,2]
[1,] 1.3640389 -0.6419722
[2,] -0.6419722 1.4642009

```

```
print(sigma_posterior_mean[, , 3])
```

```

      [,1]      [,2]
[1,] 1.0737791 0.3931505
[2,] 0.3931505 1.2466588

```

```
as.data.frame(Sigma_list)
```

```
   X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2  2.0  0.6  4.0  0.2
2 0.2 0.7  0.6  1.5  0.2  3.0
```

```
(a1 <- apply(sigma_posterior_mean, MARGIN = 3, function(aa) min(sapply(Sigma_list, function(x) sum((aa
```


[1] 0.0476229 2.7484006 0.7026725

Check identifiability issue

```
# re-run the model with wrong initial weights to check identifiability issue
set.seed(123)
fit2 <- mewishart(S_list,
  K = my_K_initial, init_pi = c(0.15, 0.40, 0.45),
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```
Iter    1 | LL=-1889.7 | 927.7 iter/sec
Iter   500 | LL=-1842.0 | 948.8 iter/sec
Iter  1000 | LL=-1839.2 | 955.5 iter/sec
Iter  1500 | LL=-1840.6 | 950.2 iter/sec
Iter  2000 | LL=-1845.6 | 954.7 iter/sec
Iter  2500 | LL=-1846.2 | 957.7 iter/sec
Iter  3000 | LL=-1839.5 | 957.7 iter/sec
Iter  3500 | LL=-1838.7 | 952.7 iter/sec
Iter  4000 | LL=-1836.5 | 948.2 iter/sec
Iter  4500 | LL=-1830.5 | 945.4 iter/sec
Iter  5000 | LL=-1831.9 | 938.9 iter/sec
Iter  5500 | LL=-1837.6 | 938.0 iter/sec
Iter  6000 | LL=-1831.5 | 936.6 iter/sec
Iter  6500 | LL=-1833.8 | 936.2 iter/sec
Iter  7000 | LL=-1836.7 | 935.9 iter/sec
Iter  7500 | LL=-1832.5 | 935.1 iter/sec
Iter  8000 | LL=-1828.3 | 933.4 iter/sec
Iter  8500 | LL=-1829.5 | 931.8 iter/sec
Iter  9000 | LL=-1829.5 | 927.9 iter/sec
Iter  9500 | LL=-1836.3 | 925.9 iter/sec
Iter 10000 | LL=-1837.3 | 922.4 iter/sec
```

```
colMeans(fit2$pi) # estimated pi
```

```
[1] 0.2548064 0.4012717 0.3439219
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit2$nu, 2, mean) # estimated nu
```

```
[1]  5.170256 12.763600  6.961533
```

```
dat$nu # true nu
```

```
[1]  8 12  3
```